Increasing field

Y14 (2014) — English translation

Inside a long solenoid of radius R, the magnetic field strength increases with time according to the linear law $B_{\text{in}} = \alpha \cdot t$. Outside, in a plane perpendicular to the axis of the solenoid and passing through its middle, we can assume that the field is $B_{\text{out}} = 0$.

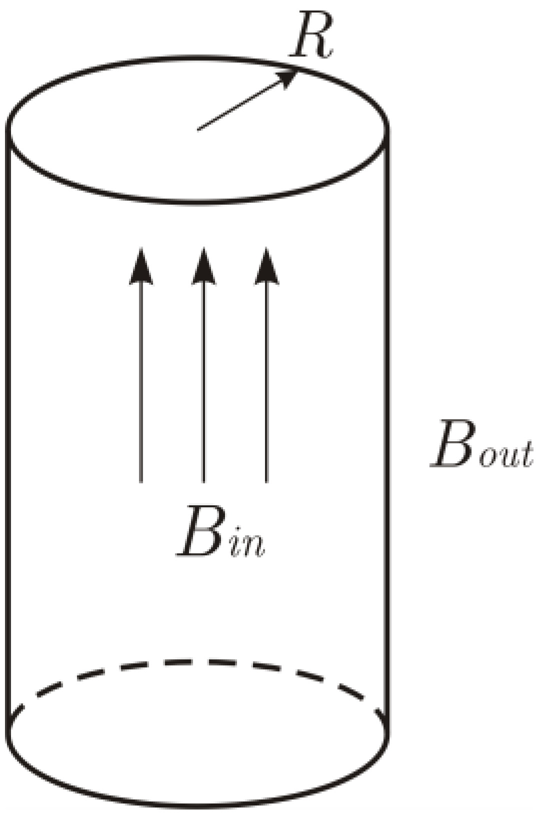

Fig. 1.

A1	Determine the electric field strength E inside and outside the solenoid as a function of r, the distance from the solenoid axis. Construct a graph of $E(r)$.	**4.00**

Source: https://pho.rs/p/3988